ALI EREN SIMSEK

Google Scholar ◇ Inspire-HEP ◇ Github ◇ LinkedIn
Phone: (+1) 215-200-8640 ◇ Email: ali.eren.simsek@cern.ch

POSITIONS HELD

L3 Convener, MC&IT Convener (Trigger) of CMS NPS (SUS/SUSY) *2024—2026*
Executive Member, Fermilab Users and Associates Executive Committee (UAEC)
- Co-Chair, Science Mission SubCommittee *2025—2026*
- Co-Chair, Government Relations SubCommittee *2024—2026*

Convener, TFPX Assembly Group - CMS Phase-II Inner Tracker *2023—2025*
Beamline Simulation Coordinator, CERN SPS H2 & H4 Beamlines *2019—2026*
Exotica EOS Manager *2026—*
Exotica HLT Scouting Contact *2026—*

SELECTED PUBLICATIONS

[1] A. E. Simsek, "Probing new physics with dedicated data streams at CMS," 2026. arXiv: 2601.13441 [hep-ex].
[2] A. E. Simsek, "RIDDLE: Revealing Irregularities through Distributional Discrepancies in Latent-space via Stein-Witness Estimation," 2026. arXiv: 2610.31370 [hep-ph].
[3] "Search for dijet resonances with data scouting in proton-proton collisions at $\sqrt{s} = 13$ TeV," 2025. DOI: 10.1007/JHEP04(2026)133. arXiv: 2510.21641.
[4] "Timing performance of the CMS High Granularity Calorimeter prototype," *JINST*, vol. 19, 2024, ISSN: 1748-0221. DOI: 10.1088/1748-0221/19/04/p04015. arXiv: 2312.14622.
[5] "Performance of the CMS High Granularity Calorimeter prototype to charged pion beams of 20-300 GeV/c," 2023. DOI: 10.1088/1748-0221/18/08/P08014. arXiv: 2211.04740 [physics.ins-det].
[6] "Response of a CMS HGCAL silicon-pad electromagnetic calorimeter prototype to 20–300 GeV positrons," *JINST*, vol. 17, 2022, ISSN: 1748-0221. DOI: 10.1088/1748-0221/17/05/p05022.
[7] "The DAQ system of the 12,000 channel CMS high granularity calorimeter prototype," *JINST*, vol. 16, 2021, ISSN: 1748-0221. DOI: 10.1088/1748-0221/16/04/t04001.
[8] "Construction and commissioning of CMS CE prototype silicon modules," *JINST*, 2020. arXiv: 2012.06336.
[9] "Search for dijet resonances using events with three jets in proton-proton collisions at $\sqrt{s} = 13$ TeV," *Physics Letters B*, vol. 805, 2020, ISSN: 0370-2693. DOI: 10.1016/j.physletb.2020.135448.
[10] "Search for high mass dijet resonances with a new background prediction method in proton-proton collisions at $\sqrt{s} = 13$ TeV," *JHEP*, 2020, ISSN: 1029-8479. DOI: 10.1007/jhep05(2020)033.

Full Publication List & Citation: *link to Inspire-HEP*

INTERNAL PAPER REVIEWS (CMS ARC/IRC)

NPS-25-015 (ARC) : Search for leptoquark pair production decaying into Charm quark and Muon
SMP-26-008 (ARC) : Search for lepton scattering in pp collisions – *"first evidence of leptons in pp collisions"*
EXO-24-039 (ARC) : Search for paired dijets with b-jets in the final state
B2G-25-004 (IRC) : Study of ZZ and ZH production in the $bb\tau\tau$ final state
HIG-24-003 (IRC) : Search for VVH(bb) production through Vector Boson Scattering
SUS-23-016 (IRC) : Search for new physics in the monophoton final state
EXO-22-026 (IRC) : Model-agnostic search for dijet resonances with anomalous jet substructure

SELECTED SEMINARS, COLLOQUIUMS, & TALKS

CMS Tracker and HL-LHC upgrade *22 June 2026*
Invited Lecture @ Fermilab USCMS Undergraduate Summer Internship (PURSUE) 2026

Fermilab Users and Affiliates Executive Committee (UAEC) Report *03 June 2026*
Invited Plenary @ USCMS Annual Meeting @ UMD

CMS Inner Tracker Forward Pixel (TFPX) Upgrade Report *05 June 2026*
Invited Plenary @ USCMS Annual Meeting @ UMD

Probing new physics with dedicated data streams at CMS *19-21 May 2025*
Phenomenology Symposium @ University of Pittsburgh (USA)

Revealing Exotic Physics through Dijet Resonance Search *11 Oct 2023*
Physics Colloquium @ Catholic University of America (USA)

Dijet Resonance Search & CMS Phase-II Upgrade *31 Mar 2022*
Seminar @ University Of Alabama (USA)

Dijet Resonance Search & CMS Phase-II Upgrade *06 Jan 2021*
Seminar @ the University Of Virginia (USA)

Dijet Resonance Search with Calo-Scouting Technique *18 Apr 2021*
APS April 2021 Meeting

FELLOWSHIPS, GRANTS & AWARDS

Breakthrough Prize - Fundamental Physics Laureates [CMS Collaboration] *2025*
Turkish Energy, Nuclear and Mining Research C. [TENMAK] Fellow - CERN *2021*
TUBITAK Fellow (2214A) - Fermilab *2020*
CMS HGCAL Group Fellow - CERN *2019*
Turkish Atomic Energy Authority [TAEK] Fellow - CERN *2018*
Turkish Atomic Energy Authority [TAEK] Fellow - CERN *2016*

PROFESSIONAL EXPERIENCE

Postdoctoral Research Assoc. at the Catholic University of America, USA *2022 –*
Graduate Researcher at CERN, Switzerland *2020-2022*
Graduate Researcher at Fermilab, USA *2019-2020*
Graduate Researcher at CERN, Switzerland *2016-2019*
Mathematics Teacher at the Ministry of Education, Turkiye *2014*
Software Developer at the 2 Adam Software & Technology, Turkiye *2009-2011*

OUTREACH ACTIVITIES

Invited Lecturer/Speaker, Fermilab PURSUE Program *2026*
Co-organizer, Annual DC Trip Organization for HEP Advocacy (UAEC at Fermilab) *2024—2026*
Facilitator/Educator, Fermilab CMS Data Analysis School (DAS) *2024*
Final Reviewer, Fermilab PURSUE Program *2024*
Ph.D. Physicist, QuarkNet Program *2022—2026*

EDUCATION

Ph.D. in Applied High-Energy and Particle Physics — Cukurova University, Turkiye *2017-2022*
- *2nd Thesis Advisor: Fermilab Senior Scientist Robert Harris*

M.Sc. in Applied High-Energy and Particle Physics — Cukurova University, Turkiye *2014-2017*

Pedagogical Formation Program in Physics Teaching — Cukurova University, Turkiye *2014*
B.Sc. in Physics — Cukurova University, Turkiye *2010-2014*
Associate Degree in Finance — Anadolu University, Turkiye *2011-2013*
Youth Software Engineering Program — Nurten Yetimoglu, Turkiye *2006-2010*

CURRENT ACTIVITIES & RESEARCHES

Development of a New Anomaly Detection Method for Particle Physics
Developed a new machine-learning anomaly detection framework based on neural Stein witnesses for model-agnostic searches for new physics. The method uses deep neural networks to identify deviations from the data without relying on predefined signal or background models. The study focuses on improving signal–background discrimination while maintaining stable and reproducible performance and minimizing artificial sculpting of the dijet mass spectrum. Benchmark studies against idealized and fully supervised reference models show that the method achieves performance nearly matching the idealized benchmark, together with robust sensitivity and minimal mass sculpting.

CMS Model-Agnostic Dijet Resonance Search with Anomaly Detection
Leading the development of a CMS model-agnostic dijet resonance search that applies my newly developed anomaly-detection method alongside R-ANODE and LaCATHODE as complementary approaches. The analysis combines anomaly-based event selection with a model-agnostic bump hunt in the dijet invariant-mass spectrum, targeting localized excesses without relying on predefined signal hypotheses. It builds on my published Run-2 dijet search (EXO-23-004), which set the most stringent CMS limits to date on low-mass dijet resonances and dark-matter mediator couplings, and aims to further extend the sensitivity to new-physics signatures using Run-3 CMS data.

Dijet Resonance Search using Run-3 PF-Scouting data at $\sqrt{s} = 13.6$ TeV 2025—
I am the official second PhD thesis advisor of a graduate student, Zohra AZIMI, whose research focuses on dijet resonance searches using Run-3 PF-Scouting data collected with the CMS Detector at $\sqrt{s} = 13.6$ TeV. My role is to guide the student from the early stages of research design and framework development to the final interpretation and presentation of the results, with a strong emphasis on rigorous data analysis in high-energy physics.

High-Mass Dijet Resonance Search with CMS data at $\sqrt{s} = 13.6$ TeV 2022—
I advised a graduate student, Doruk AGYEL SIMSEK, on her PhD thesis, which focused on high-mass dijet resonance searches in proton-proton collision data collected with the CMS Detector at $\sqrt{s} = 13.6$ TeV. My role was to guide the student from the early stages of research design and framework development to final publication, with a focus on ensuring rigorous data analysis and interpretation in high-energy physics. She successfully defended her PhD thesis and graduated in 2026. The project is still ongoing and is expected to be finalized with a publication.

CMS Inner Tracker Silicon Module Production for CMS Phase II Upgrades 2022—
Playing a central role in the production of silicon sensor modules in CUA's ISO-8 cleanroom for the Inner Tracker Forward Pixel (TFPX) detector, a key deliverable of the CMS Phase II upgrades aimed at sustaining detector performance under high-luminosity conditions. Responsibilities included coordinating day-to-day assembly workflows, enforcing quality-control procedures, and supervising a multidisciplinary team consisting of a lab technician, two graduate students, and several undergraduate interns.

PREVIOUS RESEARCH EXPERIENCE

L3 Convener, NPS MC&IT Group 2024—2026

*As the L3 Trigger Convener for the **N**ew **P**hysics with **S**tandard Objects (NPS) group, I lead the design, optimization, and validation of trigger strategies to support a wide range of new physics searches. I work in close coordination with analysis teams to ensure trigger coverage aligns with physics goals and detector constraints.*

Development of the CMS Trigger Performance Monitoring (TPM) Platform

I've developed a robust, scalable trigger monitoring system to cover all CMS PAGs (e.g., EXO, NPS, HIG, etc.) and POGs (e.g., Muon, HLT, JetMET, etc.) activities. The final goal is to run autonomously using effective ML techniques for real-time performance evaluation.

CalVision Data Acquisition (DAQ) Framework

I developed a Python-based Data Acquisition (DAQ) system from scratch for CalVision BGO crystal testing that consistently monitors and records crystal test campaigns. It streamlines how different instruments and test setups feed data into a common workflow, making it easier for students and collaborators to run measurements, document results, and compare results across studies. The framework is designed to be maintainable and expandable.

Design and Development of the University Project Management (UPM) Platform

I designed and developed a University Project Management (UPM) Platform for my current institution, The Catholic University of America, where it is now being used as an institution-wide web-based system that supports the full life cycle of sponsored grant/award applications, including proposal development, internal review workflows, sponsor requirements, award tracking, deadline management, budget oversight, and reporting. The platform integrates a Python/FastAPI backend, Oracle-backed data services, a modern JavaScript/HTML/CSS user interface, role-based administration, document-centered workflows, and AI-assisted proposal review to provide a secure, scalable, and transparent framework for principal investigators, research administrators, and university leadership.

Advancement in CMS Tracker Monitoring Tool

I contributed to the development of the CMS Tracker Data and Performance Monitoring tool by extending its capabilities and improving its usability for day-to-day detector operations. In particular, I introduced new functionalities such as real-time monitoring of auto-masked channels to assess the status of individual channels and diagnose the origin of issues quickly.

Dijet Resonance Search with the Calo-Scouting Technique Using 117 fb^{-1} at $\sqrt{s} = 13$ TeV

Performed a dijet resonance search targeting new phenomena such as dark-matter mediators, Grand Unified Theory (GUT) states, or models with extra dimensions, using the CMS calo-scouting data stream to access lower-mass regions with high statistical precision. The analysis set the most stringent CMS limits to date on the universal coupling g'_q of a leptophobic Z' boson to quarks, achieving expected sensitivities comparable to those of ATLAS and slightly stronger observed limits in the relevant mass range. The results were presented at major international conferences (Moriond 2025, Pheno 2025) and published in JHEP (Q1 journal).

SPS H4 Beamline Simulation – CMS ECAL Test-Beam

Building on the techniques developed for the HGCAL test-beam program, this study extended the beamline modeling to the SPS H4 line used for mostly CMS ECAL tests. A comprehensive treatment of material effects, Bremsstrahlung, synchrotron radiation, multiple scattering, and even environmental conditions (humidity and temperature) was included. Using data collected during in-situ visits, a detailed and realistic H4 beamline configuration was implemented, resulting in a highly accurate Monte Carlo (MC) description of the test-beam conditions and a more reliable interpretation of ECAL performance studies.

SPS H2 Beamline Simulation – HGCAL Test-Beam
This work addressed the energy loss observed in the October 2018 HGCAL test-beam data by combining on-site measurements with a high-fidelity Geant4-based simulation of the SPS H2 beamline. The study examined the impact of vacuum and air sections, as well as material interactions with beamline instrumentation such as scintillators and wire chambers. This detailed modeling improved agreement between test-beam data and Monte Carlo (MC) simulations, supported a more reliable HGCAL calibration, and prompted beamline adjustments that enabled cleaner beam delivery by the H2 experts. The same realistic beamline description was later incorporated into the FASER MC, improving its simulation accuracy by about 20%.

High Granularity Calorimeter (HGCAL) Muon/MIP Calibration with Test Beam Data
The goal of this work was to calibrate each HGCAL channel using the Minimum Ionizing Particle (MIP) constant, which maps raw ADC counts to a common MIP-based energy scale and allows the energy deposited in the calorimeter to be expressed in MIP equivalents. By suppressing noise contributions in the ADC spectra, the procedure improved the stability and uniformity of the electronic response at the module, chip, and channel levels. In addition to the standard muon-based approach, the study also showed that a comparable MIP calibration can be derived from hadronic particles (e.g., pions), providing a practical alternative when dedicated muon beams are limited.

High Granularity Calorimeter (HGCAL) Test-Beam Data Analysis
HGCAL is a new CMS sub-detector designed to measure both electromagnetic and hadronic particles. In this study, I worked on identifying and separating these two particle populations in test-beam data by constructing beam profiles from the calorimeter response. This approach also allowed us to detect and isolate performance issues in specific regions of the detector and trace the reason to nearby electrical components present during the test beam. Based on these findings, the HGCAL team decided to take corrective actions in the affected area and to relocate or shield the nearby electronics to improve the performance and stability of the impacted modules, chips, and channels in future test beams.

Troubleshooting Power Module Issues in the CMS Back-End Electronics
I worked on resolving temperature-related issues that arose in the power modules of the CMS backend electronics during the proton-proton collisions at the CMS experiment. We've developed a Raspberry Pi-based system to collect temperature readings from the modules for further analysis. The "NATview" and "Telnet" frameworks are used for management and configuration of the Schroff and Vadatech microTCA crates. Various strategies were developed and implemented to reduce the heat generated by the power modules during Run-2 data-taking! Ultimately, the HCAL group agreed on a special configuration that was applied as the solution for these temperature problems. The solution has been remarkably successful during the Run-3 data taking.

Performance and Background Noise Analysis of Hadronic Forward (HF) Calorimeter with 2016 CMS Data
During a major maintenance and upgrade period for the CMS experiment at CERN between 2013 and 2015 (Long Shutdown 1), necessary improvements were made to acquire data at an energy level of 13 TeV. In 2016, the CMS detector began collecting data from proton-proton collisions at this enhanced energy level for the first time. This project focused on analyzing the performance and stability of a particular sub-detector of the CMS Experiment called the Hadronic Forward (HF) Calorimeter, which is crucial for detecting certain types of particles. Specifically, I assessed the efficiency of noise filters in the HF Calorimeter using the data collected from proton-proton collisions at 13 TeV in 2016.

MENTORING & OUTREACH PROJECTS

Co-Chair, Science Mission Subcommittee of the UAEC 2024—2026

Serving as co-chair of the newly formed Science Mission Subcommittee (SMC) of the UAEC at Fermilab, an independent body composed of 12 members elected by the Fermilab community to represent users' and affiliates' interests in an advisory role to Fermilab management. As part of this effort, I am co-leading the establishment of the subcommittee as a standing committee by contributing to its foundational structure, including charter development, scope and responsibility definitions, and broader planning to strengthen engagement between the user community and laboratory leadership on matters related to Fermilab's scientific mission.

Organization of the Annual DC Trip for HEP Advocacy through the UAEC 2024—2026

As part of my service within the Users' Executive Advisory Committee (UAEC) at Fermilab, I co-organized the Annual Washington, DC Trip that connects members of the Fermilab community with lawmakers and policy stakeholders. This effort includes coordinating logistics, shaping the advocacy agenda, and helping participants communicate the scientific and societal value of high-energy physics. Through this work, I contribute to strengthening the connection between the scientific community and federal decision-makers who influence the future of U.S. science.

Training, Outreach, and Mentoring within the TFPX Module Production Program

In parallel with the production activities at CUA, I contributed to the onboarding and day-to-day mentoring of students and interns involved in the TFPX assembly work. I provided hands-on guidance on cleanroom practices, documentation, and basic detector concepts, enabling undergraduate and early-stage graduate students to participate in CMS hardware tasks in a structured manner. By helping define clear, achievable tasks, I supported broader student participation in the CMS Phase II activities at CUA and contributed to building a steady pool of trained personnel for ongoing module production.

Cosmic Ray Detector for Educational Outreach and Muon Lifetime Studies

I built a cosmic ray detector utilizing four photomultiplier tubes (PMTs) to identify and measure the lifetime of cosmic muons. This project served a dual purpose: to conduct high-precision measurements of muon decay and to engage high school and undergraduate students in hands-on research within high-energy physics, fostering a deeper interest in the field and cultivating the next generation of physicists.

Development of a Basic Particle Detector with Visual Tracking for Educational Outreach

I designed and 3D-modeled a compact detector system that combines Geiger tubes (LND-712) with a cloud-chamber-style visual component to detect and simultaneously display alpha, beta, and gamma particle activity. Data from the tubes were digitized with an MCP3008 ADC and read out via a Raspberry Pi, allowing students to record and analyze radiation signals while also seeing particle tracks in real time. This setup was intentionally built for high school and undergraduate outreach, giving students a tangible way to connect detector principles with actual measurements and fostering a deeper understanding of particle interactions.

Silicon Photomultiplier (SiPM) Performance Analysis of the CMS Hadronic End-Cap

I mostly mentored a graduate student to analyze the performance of SiPMs that are used in Hadronic calorimeters of the CMS Experiment. Initial studies focused on a specific part of the detector, where these new sensors were first used, and found everything to be working well. However, we spotted minor technical issues in some eta-phi regions of the detector due to a power supply failure (2018). After the investigation, the impact of this problem on the actual physics data was found to be minimal. Eventually, the problem was fixed by replacing the faulty equipment during the LS2.